A Defect in Dempster-Shafer Theory

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1 INTRODUCTION

There are four postulates accepted in D-S theory:

1. *Chance* is the limit of the proportion of "positive" outcomes among all outcomes.
2. Chance is a special case of *belief functions*.
3. *Evidence combination* refers to the pooling, or accumulating, of distinct bodies of evidence.
4. *Dempster's rule* can be used on belief functions for evidence combination.

Our results are:

- The postulates are inconsistent.
- The best solution of the inconsistency is to reject the last postulate.
- There is an approach better than D-S theory.

2 A SIMPLIFIED SITUATION

To show the inconsistency, we only need to discuss the simplest non-trivial *frame of discernment*

$$\Theta = \{H, \bar{H}\}.$$

In such a situation, all the values discussed in D-S theory over Θ can be derived from an ordered pair

$$< b,\ p >$$

where b is the *degree of belief* of $\{H\}$, and p is the *degree of plausibility* of $\{H\}$.

If $< b_1, p_1 >$ and $< b_2, p_2 >$ are based on two distinct bodies of evidence, then, according to the **Postulate 4**, the result of evidence combination is

$$\begin{aligned} b &= \frac{b_1 p_2 + b_2 p_1 - b_1 b_2}{1 - b_1(1 - p_2) - b_2(1 - p_1)} \\ p &= \frac{p_1 p_2}{1 - b_1(1 - p_2) - b_2(1 - p_1)} \end{aligned} \qquad (1)$$

To specify the meaning of "evidence combination", Shafer introduces *weight of evidence*, w, with the following properties (the **Postulate 3**):

1. w is a measurement defined on bodies of evidence, with respect to a subset of Θ, and it takes values on $[0, \infty]$.

2. When two entirely distinct bodies of evidence are combined, the weight of the pooled evidence (for the same subset of Θ) is the *sum* of the original ones.

Therefore, if we use w^+ and w^- to indicate the weight of evidence for $\{H\}$ and $\{\bar{H}\}$, respectively, the combination rule must satisfy

$$\begin{aligned} w^+ &= w_1^+ + w_2^+ \\ w^- &= w_1^- + w_2^- \end{aligned} \tag{2}$$

Because belief functions are based on available evidence, b and p are determined by w^+ and w^-, and their relationship can be derived from (1) and (2):

$$\begin{aligned} b &= \frac{e^{w^+} - 1}{e^{w^+} + e^{w^-} - 1} \\ p &= \frac{e^{w^+}}{e^{w^+} + e^{w^-} - 1} \end{aligned} \tag{3}$$

Generally, there is $b \leq p$. When $b = p$, the belief function Bel becomes a probability function in the sense that $Bel(\{H\}) + Bel(\{\bar{H}\}) = 1$. Such a function is called "Bayesian" by Shafer, and written as Bel_∞. From (3), it is clear that this happens if and only if the sum of w^+ and w^-, w, goes to infinite:

$$Bel_\infty(\{H\}) = \lim_{w \to \infty} b = \lim_{w \to \infty} p \tag{4}$$

According to the **Postulate 2**, $Bel_\infty(\{H\})$ is identical with $q(H)$, which is the *chance*, or *(aleatory) probability*, of H. Therefore, probability functions are a subset of belief functions.

3 A PROBLEM

To see clearly how D-S theory and probability theory are related to each other, consider the situation where evidence for Θ is in the form of a sequence of experiment outcomes with the following properties:

1. No single outcome can completely confirm or refute H.
2. There are only two possible outcomes: one supports H, while the other supports $\bar{H}$.
3. The outcomes are independent and provide distinct bodies of evidence.

Therefore, both D-S theory and probability theory can be directly applied in the situation.

In the following, t^+ is the number of outcomes that support H, t^- is the number of outcomes that support $\bar{H}$, and t is the total number of outcomes ($t = t^+ + t^-$).

Because there are only two types of evidence, we can assign positive real numbers w_0^+ and w_0^-, as weight of evidence, to each outcome supporting H and $\bar{H}$, respectively. After t outcomes are observed, the weight of available evidence (for $\{H\}$ and $\{\bar{H}\}$) can be calculated according to the **Postulate 3**:

$$\begin{aligned} w^+ &= w_0^+ t^+ \\ w^- &= w_0^- t^- \end{aligned} \tag{5}$$

According to the **Postulate 1**, if the chance of H exists, it is given by

$$q = \lim_{t \to \infty} \frac{t^+}{t} \tag{6}$$

However, from (3), (4), (5) and (6), we get

$$\begin{aligned} Bel_\infty(\{H\}) &= \lim_{w\to\infty} \frac{e^{w^+}-1}{e^{w^+}+e^{w^-}-1} \\ &= \lim_{t\to\infty} \frac{e^{w_0^+ t^+}-1}{e^{w_0^+ t^+}+e^{w_0^- t^-}-1} \\ &= \begin{cases} 0 & \text{if } w_0^+ q < w_0^-(1-q) \\ 1 & \text{if } w_0^+ q > w_0^-(1-q) \\ \frac{1}{1+e^\Delta} & \text{if } w_0^+ q = w_0^-(1-q) \end{cases} \end{aligned} \quad (7)$$

where $\Delta = \lim_{t\to\infty}(w_0^- t^- - w_0^+ t^+)$.

Therefore, in contrary with the **Postulate 2**, $Bel_\infty(\{H\})$ is usually different from $q(H)$, unless $q(H)$ happens to be 0, 1, or $\frac{w_0^-}{w_0^+ + w_0^-}$, and in the last case $w_0^- t^- - w_0^+ t^+$ must have a limit Δ that makes $\frac{1}{1+e^\Delta} = q(H)$.

This inconsistency is derived from the four postulates alone, so it is independent from other controversial issues about D-S theory, such as the interpretation of the belief function, the accurate definition of *distinct* (or *independent*) bodies of evidence, and the actual measurement of the weight of evidence. No matter what opinions are accepted on these issues, as long as they are held consistently, the previous problem remains. For example, the choice of w_0^+ and w_0^- can only determine which chance value is mapped to the degree of belief $\frac{1}{1+e^{\Delta}}$ (so all the other values are mapped to 0 or 1 correspondingly), but cannot change the result that the chance and the Bayesian belief function of H are usually different.

4 POSSIBLE SOLUTIONS

To get a consistent theory, at least one of the four postulates must be removed.

If the **Postulate 1** is rejected, and "chance" is redefined as $\lim_{w\to\infty} \frac{e^{w^+}-1}{e^{w^+}+e^{w^-}-1}$, the theory will become consistent. However, this will violate the common usage of the concept "chance", and make probability theory inapplicable.

If the **Postulate 2** is rejected, probability functions are no longer special cases of belief functions. D-S theory will lose its strong relationship with probability theory, and become hard to be understood and used.

If the **Postulate 3** is rejected, and "evidence combination" is redefined as corresponding to the *multiplication*, not *addition*, of weight of evidence, the theory will become consistent. However, this will conflict with the "accumulating" and "pooling" intuition behind the notion "evidence combination".

If the **Postulate 4** is rejected, we are no longer working within D-S theory.

5 AN ALTERNATIVE APPROACH

In the *Non-Axiomatic Reasoning System (NARS)* project, a new approach is introduced, in which the **Postulate 1**, **2**, and **3** are preserved, but the "belief functions" have a different relationship with weight of evidence, comparing with D-S theory.

With the *frame of discernment* and *weight of evidence* defined as before and $w_0^+ = w_0^- = 1$, the belief about H is represented by a *lower frequency*, l, and a *upper frequency*, u, which are defined as

$$\begin{aligned} l &= \frac{w^+}{w+1} \\ u &= \frac{w^+ + 1}{w+1} \end{aligned} \tag{8}$$

Because the current frequency of positive evidence is $\frac{w^+}{w}$, after combining the current evidence (whose weight is w) with the new evidence provided by a new outcome (whose weight is 1), the new frequency will be in the interval $[l, u]$.

The new measurement of uncertainty has the following properties:

1. $l \leq \frac{w^+}{w} \leq u$, that is, the current frequency is within the $[l, u]$ interval.

2. The bounds of frequency are defined in terms of available evidence, which is finite. Whether the frequence of positive evidence really has a limit does not matter.

3. $\lim_{w\to\infty} l = \lim_{w\to\infty} \frac{w^+}{w} = \lim_{w\to\infty} u = q$. If the frequency does have a limit q, then q is also the limit of l and u.

4. However, q, if it exists, is not necessarily in the interval all the time. l and u are not bounds of the chance, or probability, of H.

5. The width of the interval $i = u - l = \frac{1}{w+1}$ monotonically decreases during the accumulating of evidence, and so can be used to represent the system's "degree of ignorance" (about the frequency).

From the definitions of the lower-upper frequencies and the **Postulate 3**, a *combination rule*, from $[l_1, u_1] \times [l_2, u_2]$ to $[l, u]$, is uniquely determined in terms of lower-upper frequencies, when neither $i_1 = u_1 - l_1$ nor $i_2 = u_2 - l_2$ is 0:

$$\begin{aligned} l &= \frac{l_1 i_2 + l_2 i_1}{i_1 + i_2 - i_1 i_2} \\ u &= \frac{l_1 i_2 + l_2 i_1 + i_1 i_2}{i_1 + i_2 - i_1 i_2} \end{aligned} \tag{9}$$

6 CONCLUSION

D-S theory establishes an unnatural relation between degree of belief and weight of evidence by using Dempster's rule for evidence combination. As a result, the assertion that "probability is a special belief function" is in conflict with the definitions of "probability" and "evidence combination."

The inconsistency is solvable within D-S theory, but such a solution will make D-S theory either lose its naturalness (by using a concept in a unusual way), or miss its original goals (by being unable to represent probability or to combine evidence).

The "lower-upper frequency" approach sets up a more natural relationship among the various measurements of uncertainty. It extends probability theory by introducing an evidence combination rule, but does not suffer from the inconsistency.

Appendix: Detailed Derivation of (6)

1. If $w_0^+ q < w_0^-(1-q)$, then

$$\begin{aligned}
\lim_{t\to\infty} \frac{e^{w_0^+ t^+} - 1}{e^{w_0^+ t^+} + e^{w_0^- t^-} - 1} &= \lim_{t\to\infty} \frac{e^{w_0^+ t^+ - w_0^- t^-} - e^{-w_0^- t^-}}{e^{w_0^+ t^+ - w_0^- t^-} + 1 - e^{-w_0^- t^-}} \\
&= \lim_{t\to\infty} \frac{e^{-t[w_0^-(1-\frac{t^+}{t}) - w_0^+ \frac{t^+}{t}]} - e^{-t[w_0^-(1-\frac{t^+}{t})]}}{e^{-t[w_0^-(1-\frac{t^+}{t}) - w_0^+ \frac{t^+}{t}]} + 1 - e^{-t[w_0^-(1-\frac{t^+}{t})]}} \\
&= \frac{0-0}{0+1-0} = 0
\end{aligned}$$

2. If $w_0^+ q > w_0^-(1-q)$, then

$$\begin{aligned}
\lim_{t\to\infty} \frac{e^{w_0^+ t^+} - 1}{e^{w_0^+ t^+} + e^{w_0^- t^-} - 1} &= \lim_{t\to\infty} \frac{1 - e^{-w_0^+ t^+}}{1 + e^{w_0^- t^- - w_0^+ t^+} - e^{-w_0^+ t^+}} \\
&= \lim_{t\to\infty} \frac{1 - e^{-t[w_0^+ \frac{t^+}{t}]}}{1 + e^{-t[w_0^+ \frac{t^+}{t} - w_0^-(1-\frac{t^+}{t})]} - e^{-t[w_0^+ \frac{t^+}{t}]}} \\
&= \frac{1-0}{1+0-0} = 1
\end{aligned}$$

3. If $w_0^+ q = w_0^-(1-q)$, and $\lim_{t\to\infty}(w_0^- t^- - w_0^+ t^+) = \Delta$, then

$$\begin{aligned}
\lim_{t\to\infty} \frac{e^{w_0^+ t^+} - 1}{e^{w_0^+ t^+} + e^{w_0^- t^-} - 1} &= \lim_{t\to\infty} \frac{1 - e^{-w_0^+ t^+}}{1 + e^{w_0^- t^- - w_0^+ t^+} - e^{-w_0^+ t^+}} \\
&= \frac{1}{1+e^{\Delta}}
\end{aligned}$$